GEK-8198

DRIVE SYSTEM INSTRUCTIONS

IC4962 TCR INDUCTOR DYNAMOMETER CONTROL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

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IC4962 TCR INDUCTOR DYNAMOMETER CONTROL

Description

The General Electric inductor dynamometer is an eddy current machine used only to absorb power.

The inductor-dynamometer consists principally of a smooth cylindrical rotor, a stator having projecting teeth and a toroidol DC field coil. Power input to the shaft is changed into eddy current losses in the rotor and the resultant heat is dissipated by cooling water. Power absorption is controlled by controlling the DC current in the dynamometer field.

A representation of the complete system and the corresponding block diagram are shown below. The dashed lines in the block diagram are for true torque regulation when supplied.

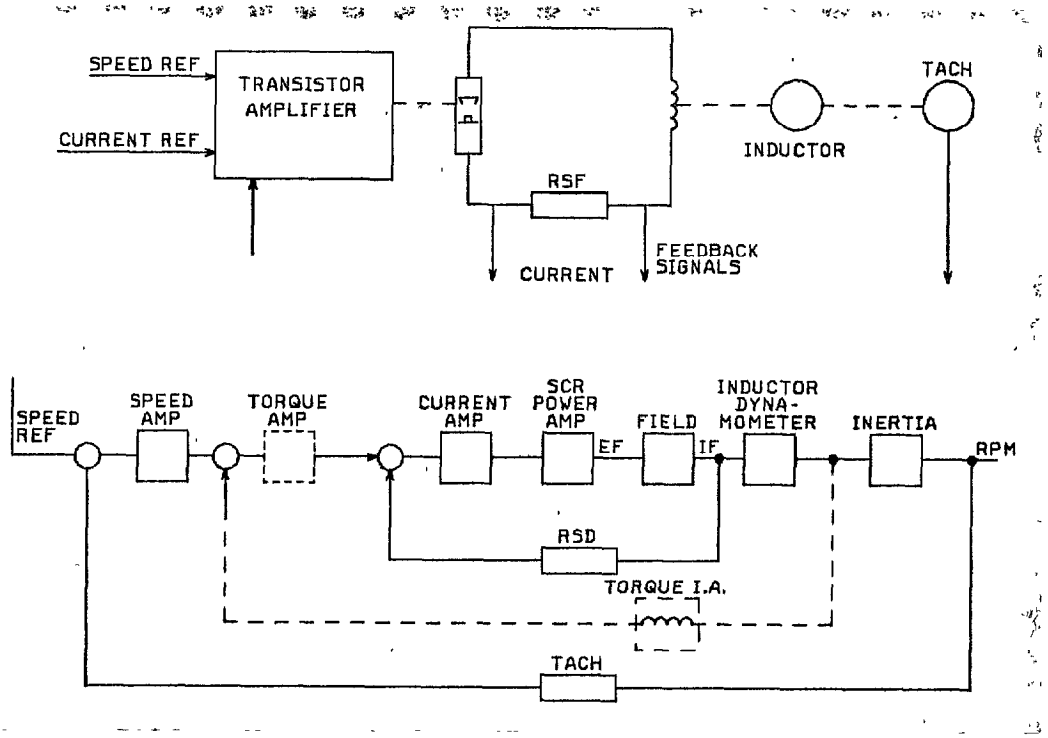


Figure 1

Adjustment

The proper procedure for adjusting the power supply, stability circuits, maximum speed, maximum current, dynamometer field current feedback, overspeed, current limit, braking current, and residual suppression are described later.

Power Supply

A regulated power supply is furnished in order to supply constant reference signals --minimizing the drift of the dynamometer operation. See IC3601A104A and IC3601A105A for details.

Speed Control

The system is capable of closely maintaining speed with 80 percent load torque change as long as the load under any condition is not less than 20 percent at the specified speed after transients have decayed to zero. A DC tachometer-generator, coupled to the dynamometer, supplies a voltage directly proportional to speed. This voltage is compared with a voltage reference set by a speed adjustment potentiometer on the operator's panel and connected to the regulated reference supply as shown in Figure 2.

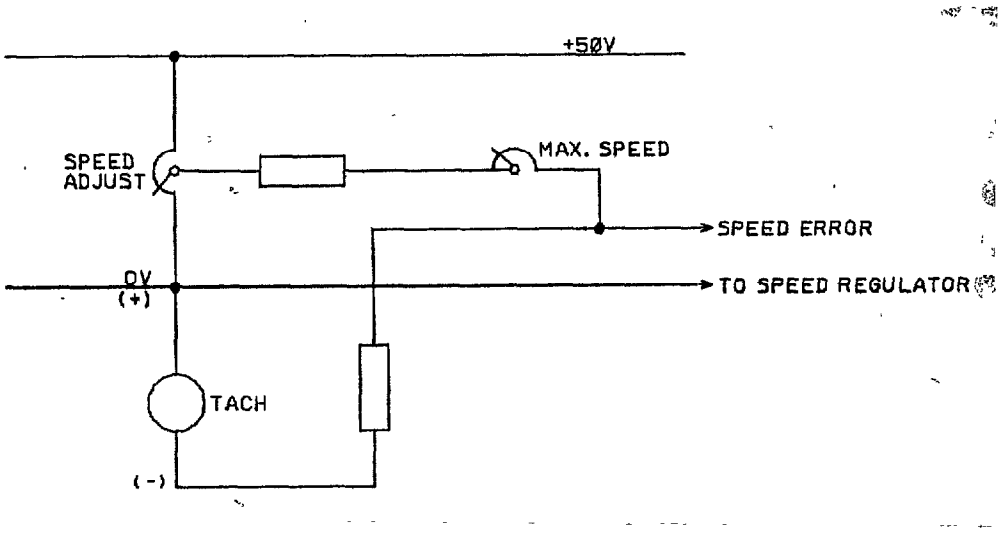


Figure 2
Speed Error Circuit

If the speed signal does not agree with the reference, the resultant error voltage will be amplified by the preamplifier (See Figure 3) which supplies a corrective signal to the current amplifier.

The SCR excites the dynamometer field which changes the applied torque in a direction to rebalance the tachometer-generator and reference signals.

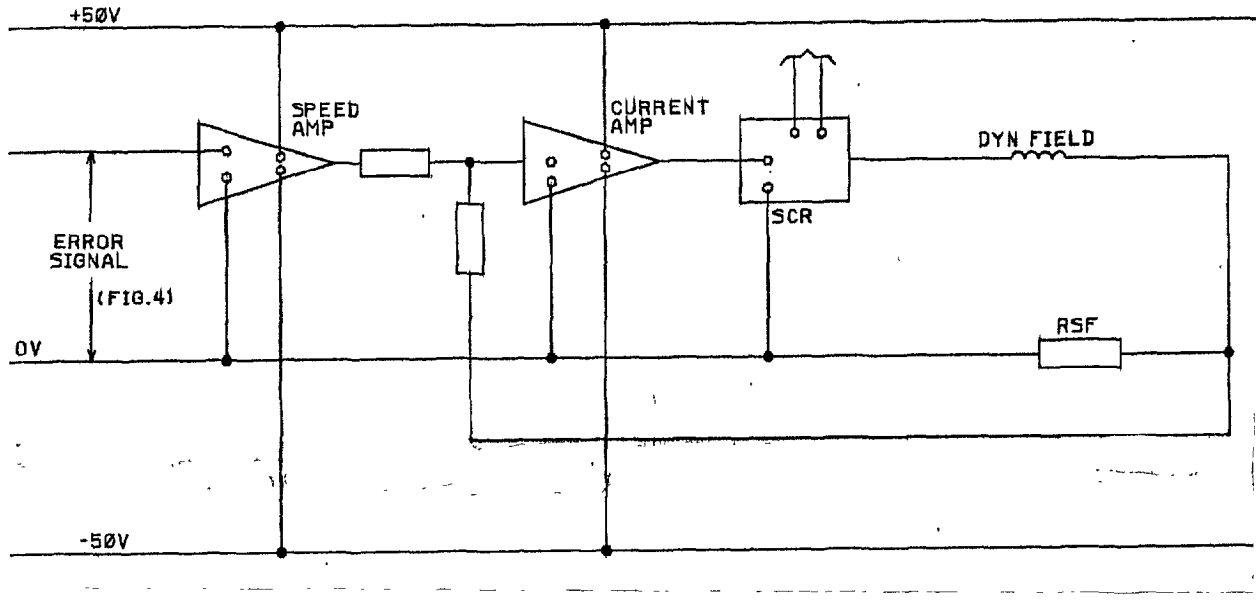


Figure 3

Field Current Regulation

A control switch on the operator's panel can be used to select speed mode or current mode. In a current mode of regulation the speed reference and tach signal is shorted to common and the desired current reference is set by the current adjustment potentiometer. This reference is compared to a signal obtained from RSF resistor in series with the dynamometer field. Any error is amplified by the current preamplifier and then applied to the SCR exciter. The SCR exciter changes the field current as necessary to hold the error at a small value. (See Figure 4).

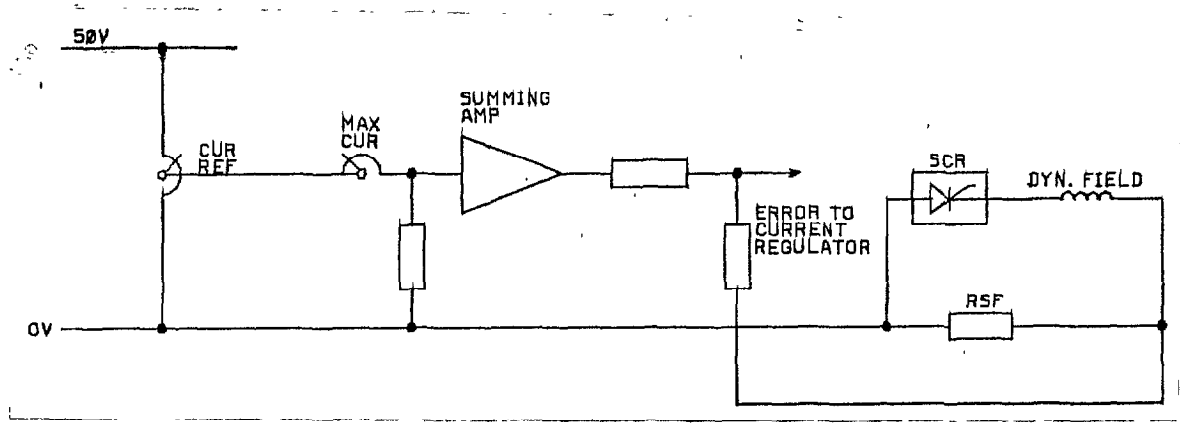
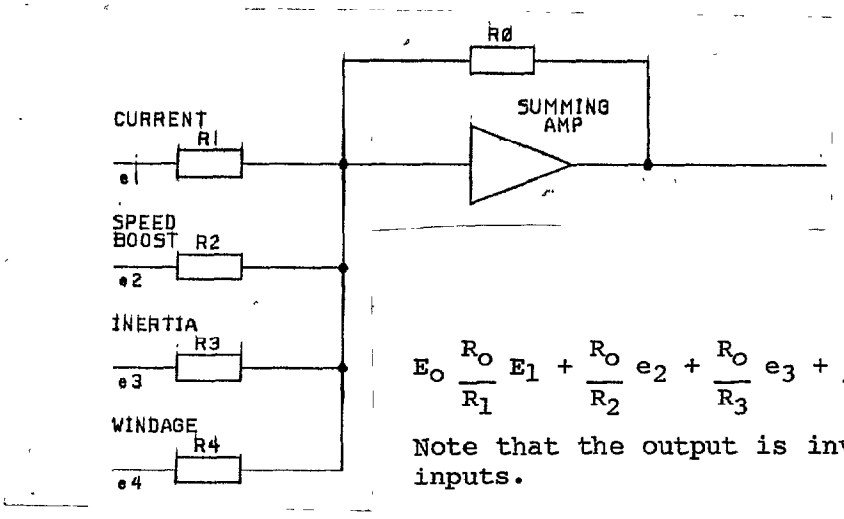


Figure 4

The speed preamplifier is used as a summing amplifier whose sole function is to assemble signals from their respective inputs and pass them through the current limit circuit into the current pre-amplifier. Operational amplifiers have the following characteristic when used for summing signals. The output of the summing amplifier is the current reference signal.



$$E_o \frac{R_o}{R_1} E_1 + \frac{R_o}{R_2} e_2 + \frac{R_o}{R_3} e_3 + \frac{R_o}{R_4} e_4$$

Note that the output is inverted from the inputs.

Figure 5

True Torque Regulation

If true torque regulation is furnished, a control switch on the operator's panel can be used to select the speed mode or torque mode. In the torque mode, the speed reference and tach signal is shorted to common and the desired torque reference is set by the torque adjustment pot. This reference is compared with a torque signal which is usually obtained from a strain gauge on the dynamometer. This is a low level signal (millivolt range) and must be amplified to the regulator voltage level. This requires a high gain amplifier.

Any noise pickup by the torque feedback circuit would be amplified by the feedback amplifier and cause poor regulator performance. For this reason, an isolation amplifier is used.

Any error between the torque reference and the feedback is amplified by the torque and current preamplifiers and then applied to the SCR exciter. The SCR exciter changes the field current as necessary to hold the error to a small value.

The speed preamplifier is used as a summing amplifier whose sole function is to assemble signals and pass them through the torque limit ciruit into the torque pre-amplifier (See Figure 5).

The torque regulator is similar to the speed regulator in that it has an inner field current loop. The output of the torque amplifier is the field current reference.

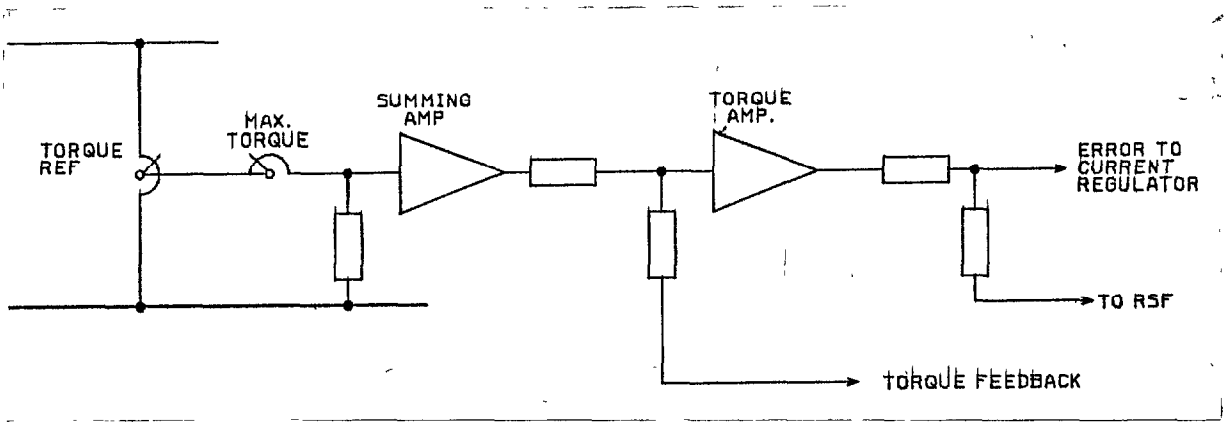


Figure 5A

Speed Boost

The operator's panel also contains a speed boost potentiometer (See Figure 6). Making adjustment here allows the operator to add a signal proportional to speed in the input circuit while under current control (torque control if available). This will provide a rising absorption-torque characteristic as the speed is increased. In the case of an engine which has a flat region in its speed-torque characteristic, this speed boost signal will allow stable operation of the dynamometer system and the engine.

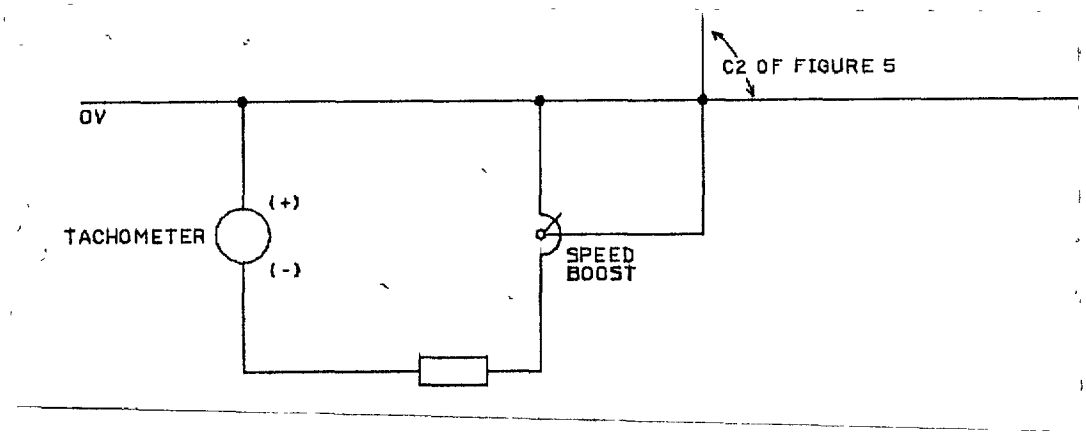
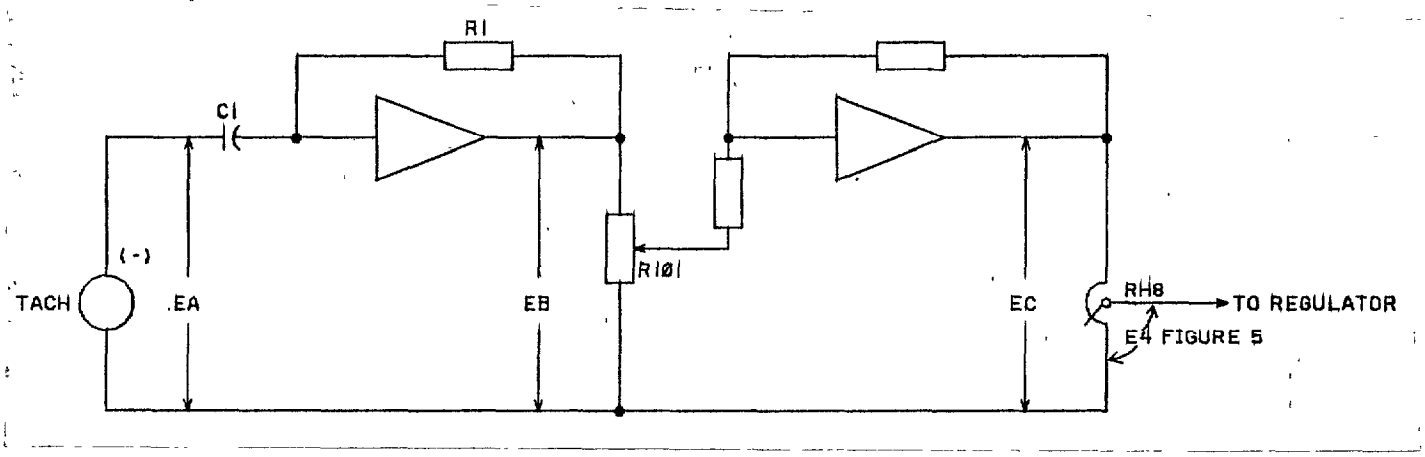


Figure 6

Electrically Simulated Inertia

To electrically simulate an inertia load on a test motor, a load must be applied proportional to the rate of change of speed. This is done by operating in a torque or current mode and adding to the reference an absorption reference proportional to the rate of change of speed. This simulates a positive inertia. See the following circuit.



If the dynamometer is accelerating, a current will flow in C1 as shown below:

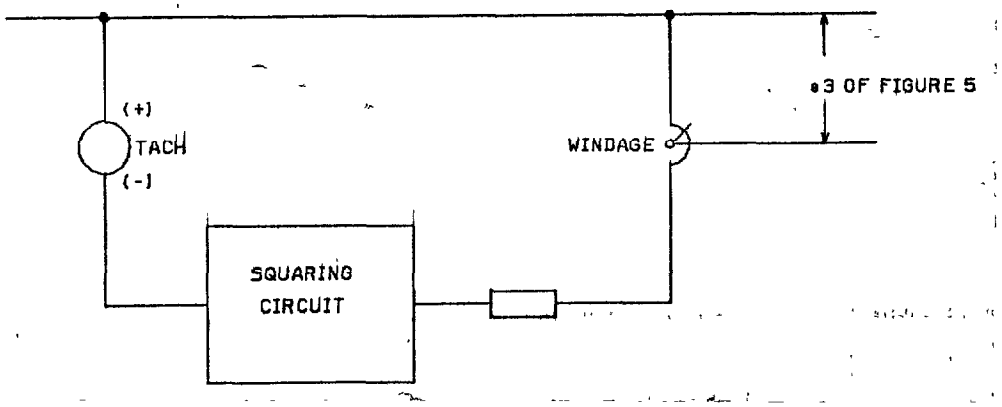
$$i = \frac{EA}{t} \quad (C1), \quad E_B = iR_1 = \frac{EA}{t} \quad (C1) \quad (R1)$$

If the acceleration rate is constant E_B and E_C are constant. A constant reference is supplied to the current regulator. The maximum reference (E_C) may be adjusted with R101 when RH8 is fully CW.

With the electrical inertia simulation, any change in throttle position on the engine will produce an opposing torque in the dynamometer. This torque will oppose the change in the engine torque such that the engine will change speed at a rate which is slower than if inertia simulation were not present. In this manner the engine and dynamometer inertia can be made to appear larger.

Windage

The windage function, change in torque or field current as the square of speed, is also included in many drives. See the following circuit.



This circuit simulates the windage force that is exerted on a vehicle.

Current Limit Circuit

Maintaining speed within close limits requires a high gain system. This means that a small error between the tachometer generator voltage and the reference voltage will cause a large change in dynamometer field voltage. This large change could cause the field current to become excessive. Therefore, a circuit which limits field current is included to protect the machinery. This is accomplished by limiting the output of the speed preamplifier. (See Figure 7). The error necessary to move the speed preamplifier is very small, therefore its potential can be considered as being zero volts at all times.

If the speed error causes the output of the speed preamplifier to exceed the voltage set by current limit potentiometer, the diode shown in Figure 7 will conduct to limit the preamplifier output to these set values, thus limiting the current. These circuits are adjusted to operate between 100 and 125 percent of rated field current.

If true torque regulation is available, the limit circuit around the speed preamplifier is a torque limit. The current limit circuit is now around the torque amplifier (see Figure 5A) and operates in the same manner as described above.

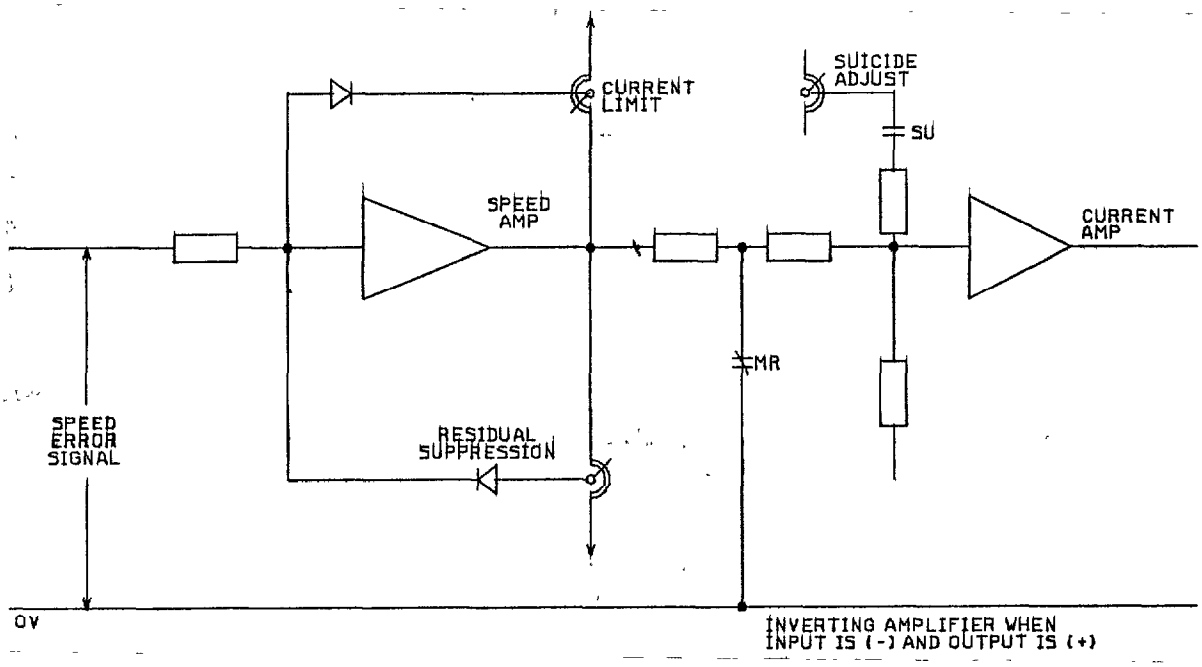


Figure 7
Current Limit and Residual Suppression Circuit

Residual Suppression

This function provides for suppression of dynamometer flux to permit operation at a minimum torque. A silicon diode rectifier supplies approximately 10 percent negative field current for residual buckout. It is adjustable (See Figure 7) from approximately -10 percent to zero.

When the drive is coast stopped, the MR relay is de-energized. The drive coast stops and at a low speed, the SU relay is energized to call for a small negative field current. It is adjusted with the suicide adjustment for minimum dynamometer torque.

Stabilizing Circuit

For good speed regulation, the system must have a high gain in order to obtain the accuracy desired. In high-gain systems where a feedback is compared with a reference signal instability may become a problem.

Instability is caused by energy storage devices and components within the closed loop system - in particular the inertia of the dynamometer and test engines, but also the field inductance and SCR inductance.

Stabilization is accomplished by use of a lag-lead or "notch" circuit (see Figure 8), which is the equivalent of the simple resistor-capacitor combination shown in Figure 9. Stabilization networks are shown later.

This circuit alters the effect of input signals according to the magnitude of their rate of change with respect to time. Proper settings of the magnitudes of C and R and adjustments on the potentiometer will produce the desired speed regulator characteristics.

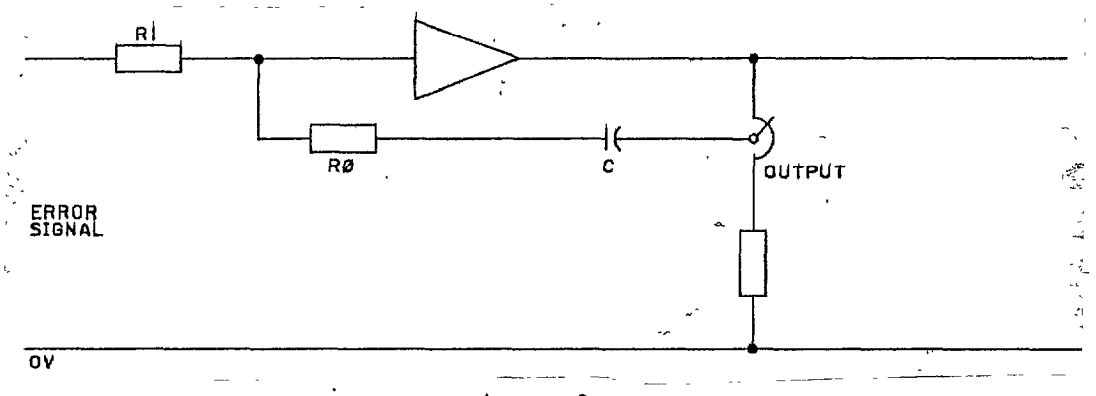


Figure 8

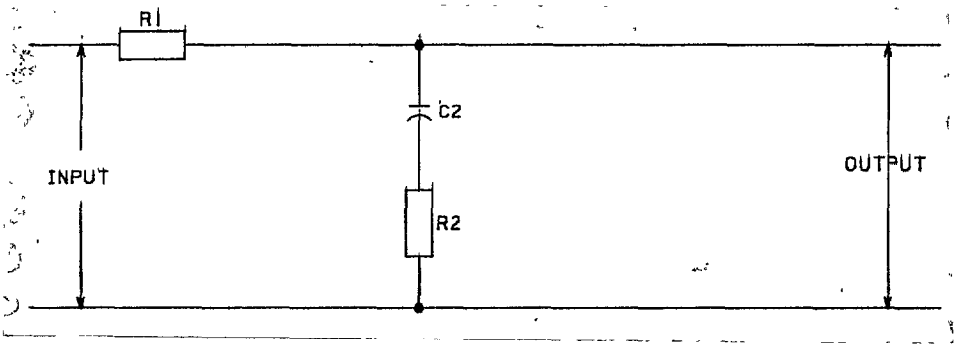


Figure 9

Emergency Stop

If the EMERGENCY STOP pushbutton is pressed, the control will automatically transfer to speed control with zero speed reference. This will cause the regulator to operate from a speed signal modified only by current limit. The dynamometer will thereby be stopped rapidly by braking. User's protective devices, such as bearing temperature relays, cooling water pressure switches, etc., can be connected in to give an emergency stop. An ignition relay will also be dropped out, and will interrupt ignition for the test engine. During braking, the current limit circuit functions to prevent a field circuit overcurrent. When the dynamometer speed gets to a low value, the dynamometer is transferred to residual suppression. To remove field to the dynamometer after an E stop, the reset pushbutton is depressed.

Adjustable Emer-Stop Field Current

This function provides an adjustable preset field current during an emergency stop to permit limiting maximum braking torque. A single turn rheostat is provided on the operator's control panel for field current adjustment. Range of adjustment -0 to rated base speed field current. (See Figure 10).

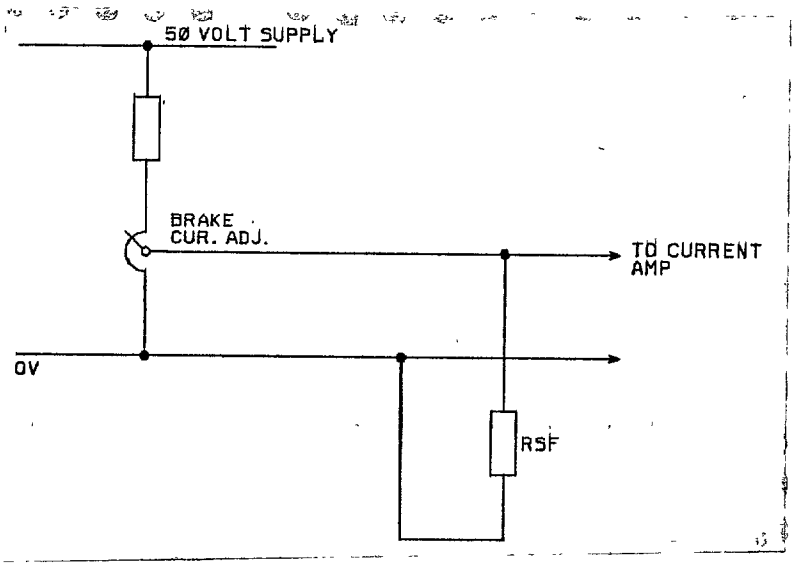


Figure 10

Protective Devices

Included in the control system are various protective devices in addition to the EMERGENCY STOP pushbutton discussed above.

1. Dynamometer overspeed (voltage sensitive relay operating from tachometer-generator)

2. Dynamometer Field Thermal Protection

3. Stop Pushbutton

4. Auxiliary Starters

Operating Instructions

The sequence of operation is outlined below:

1. Turn the transfer switch to either speed or current (or torque). This determines the mode of operation of the dynamometer. Press the auxiliary start pushbutton. The dynamometer system auxiliaries (including the oil pump motors) will start in sequence. These auxiliaries are protected with overload relays.

2. Turn the speed control rheostat to maximum (or to a value higher than the starting speed of the engine).

3. Push the reset pushbutton. This energizes the emergency stop and ignition relays.

4. Push the start pushbutton.

5. The test engine can now be energized and brought up to speed if it is not already running.

6. The speed control potentiometer can now be adjusted to obtain the desired engine speed and loading.

7. During any changes, the operator should watch the RPM indicator; voltmeter, torquemeter and ammeter in order to check the operation of the system.

8. For normal stopping of the system, the load should be removed from the dynamometer and the prime mover de-energized. The dynamometer stop pushbutton can then be pressed to de-energize the control and allow the dynamometer to coast stop.

9. For emergency stopping. Press the emergency stop pushbutton. This will rapidly bring the drive to rest under preset braking torque. The reset pushbutton must then be depressed before the drive can be restarted.

An indicating light tells the operator when the drive is ready to run. The light will not come on until all protective devices are in their normal operating condition. Further, some systems are supplied with additional indicator lights or annunciator drops to indicate that a protective device (as described above) has operated.

Operational Techniques Applied to Regulating Systems

The term operational imposes several conditions on any amplifier it is applied to. These are:

1. Gain is sufficiently large that input voltage and current are considered to be zero.

2. There is no internal time constant, i.e., bandwidth on frequency is very wide and rise time on step input from 5% to 95% of output is negligibly small.

The above conditions allow the following conclusions for the configuration shown.

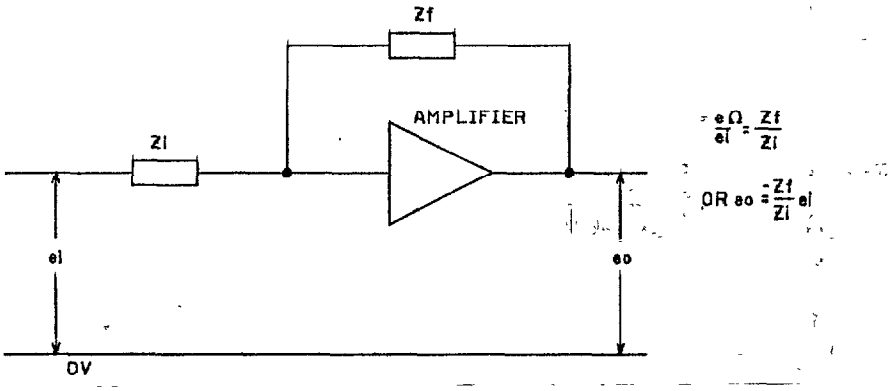


Figure 11

Z_i and Z_f can be any combination of capacitance, inductance, and resistance. The negative sign indicates 180 degree phase shift (polarity change) from input to output.

The following elementary circuits are commonly used.

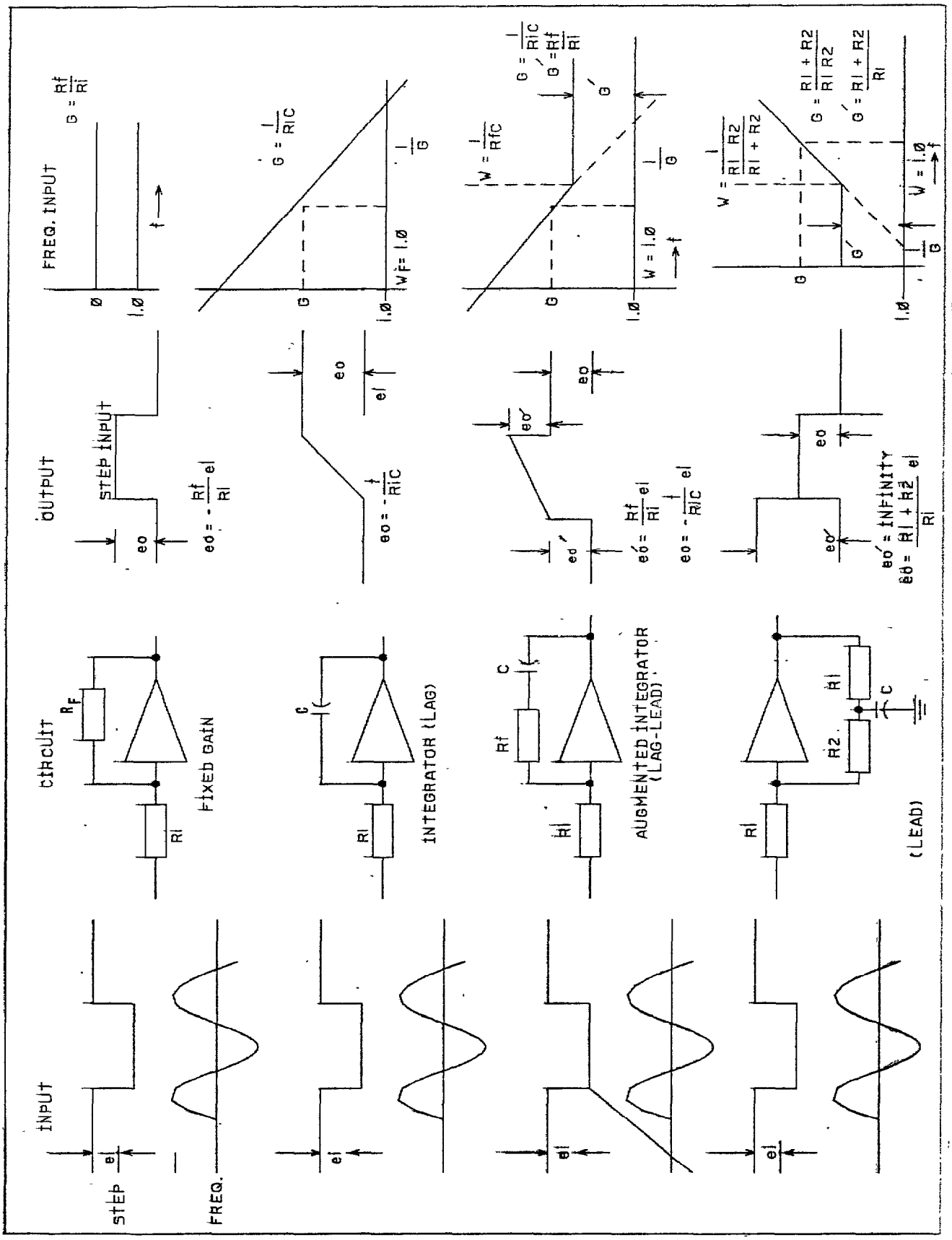

INPUT
STEP
FREQ.
CIRCUIT
FIXED GAIN
INTEGRATOR (LAG)
AUGMENTED INTEGRATOR (LAG-LEAD)
(LEAD)
OUTPUT
STEP INPUT
FREQ. INPUT

Figure 12

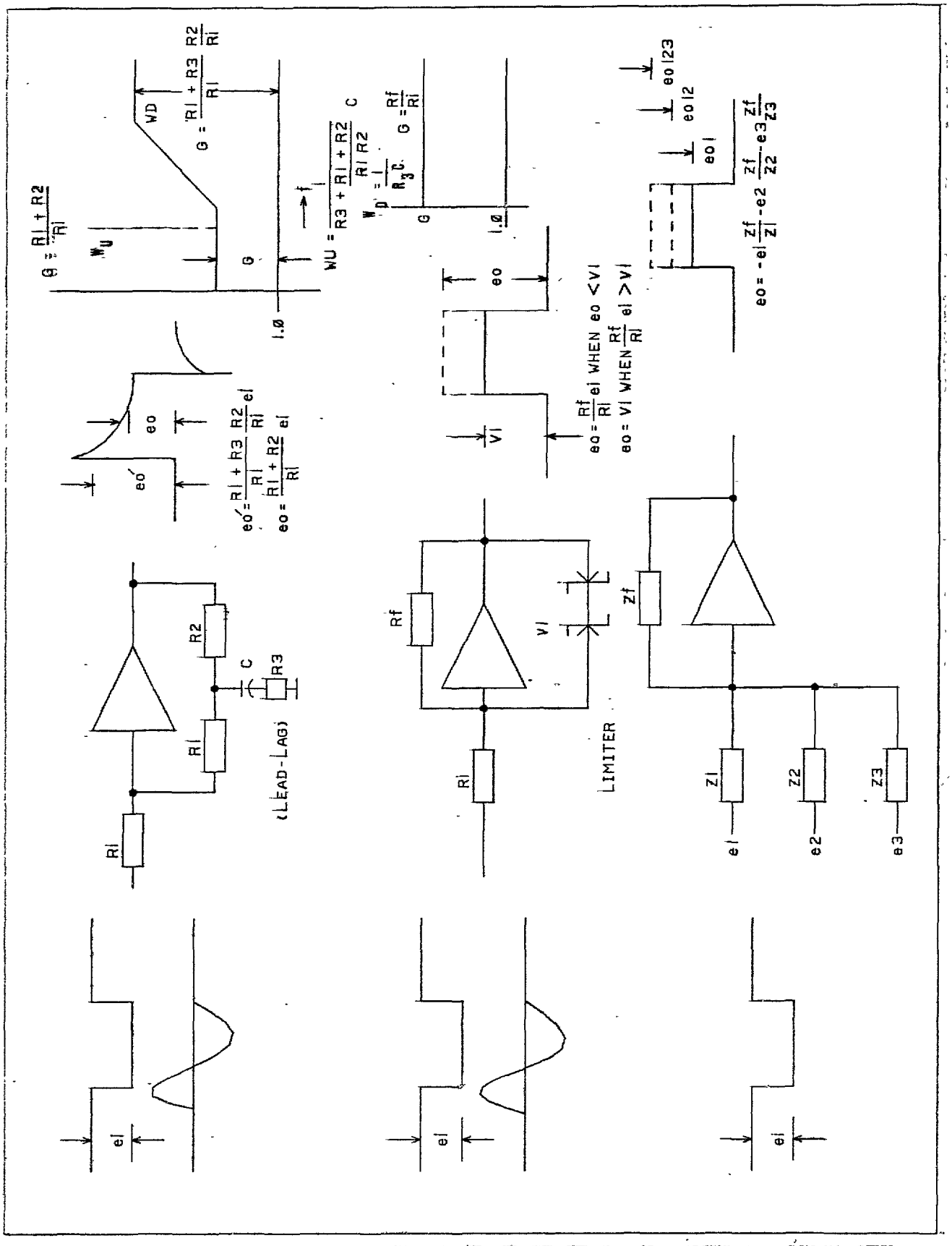

G = R1 + R2 / R1
WU
G
1.0
WD
G = R1 + R3 / R1 R2 / R1
WU = 1 / R3 + R1 + R2 / R1 R2 C
WD = 1 / R3 C
G = Rf / R1
eo
eo'
eo' = R1 + R3 / R1 R2 / R1 ei
eo = R1 + R2 / R1 ei
VI
eo = Rf / R1 ei WHEN eo < VI
eo = VI WHEN Rf / R1 ei > VI
eo123
eo12
eo1
eo = -ei Zf / Z1 - e2 Zf / Z2 - e3 Zf / Z3
R2
C
R3
R1
R1
(LEAD-LAG)
Rf
VI
R1
LIMITER
Zf
Z1
Z2
Z3
e1
e2
e3
ei
ei
ei

Figure 13

Regulating System Notes

In any regulated system using negative feedback, the phase shift from input to output must be less than 180°. Any device which has capacitance or inductance causes phase shift and attenuation. The corner frequency at which attenuation begins is the reciprocal of the time constant. In a generator field, attenuation would begin at:

$$\frac{R_f}{L_f} = \frac{1}{T_f} \quad \text{RADIANS PER SECOND}$$

If more than one time constant exists in a regulated system, stabilizing means must be added. Stabilizing normally consists of increasing the gain of the regulator in proportion to the decrease in gain (increase in attenuation) of the other system elements. Then additional controlled attenuation is added to the regulator to control the speed of response of the system.

In general (rule of thumb), response in radians per second is equal to $\frac{3}{t}$ for the following condition. (See Figure 14).

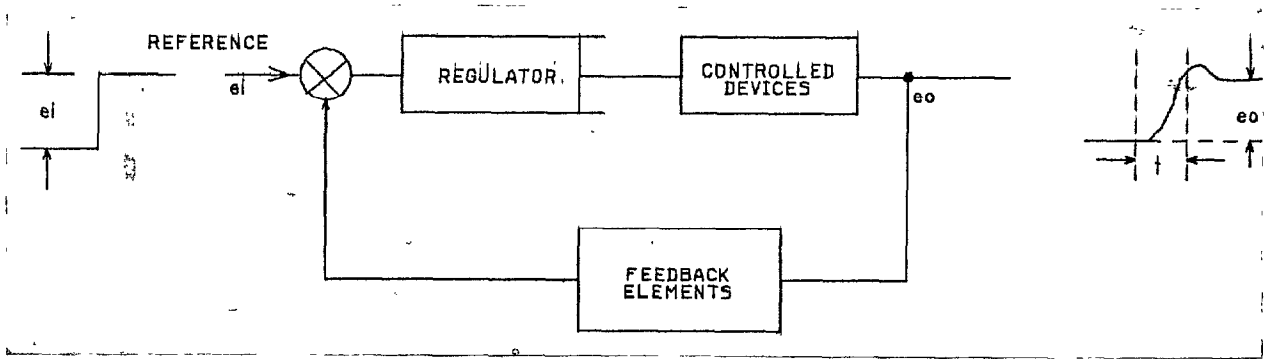


Figure 14

For the general system, any time constant lower than the response frequency or less than 3 times greater than the response frequency must be compensated. The following simple example will illustrate.. (Figure 15)

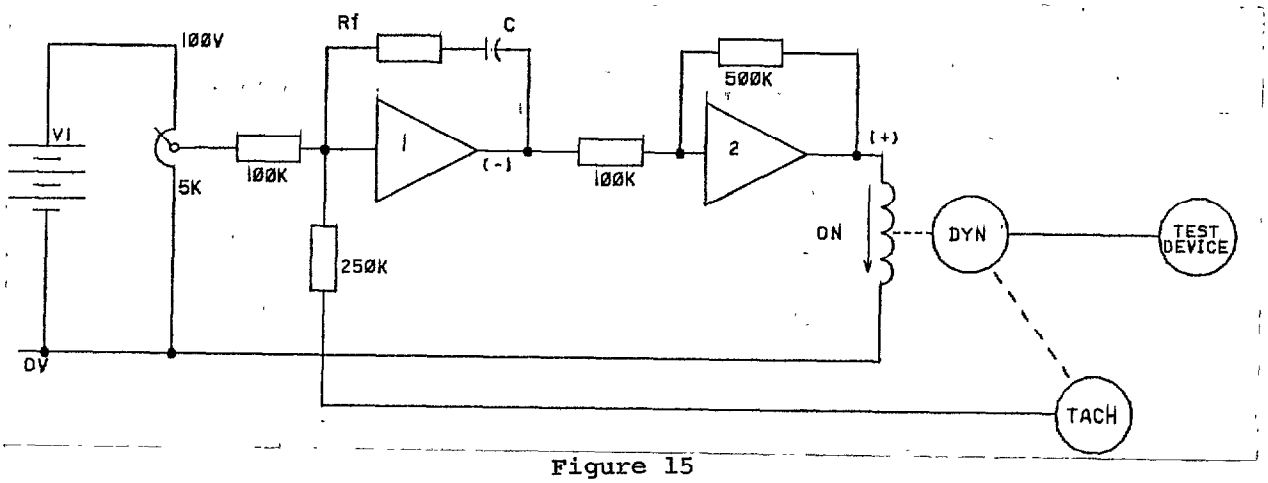


Figure 15

Assume the following:

1. No saturation occurs in any element.
2. Dynamometer field time constant = 1 second.
3. Desired response frequency = 10 radians/second.
4. Dynamometer gain = 5 volts/volt; i.e., 1 volt on field produces minus 5 volts tach.

The DC accuracy of the above system is theoretically infinite; i.e.: regardless of load, the output voltage will not change if the reference is not changed. The following conditions should exist to fit the foregoing statements.

1. R_fC = 1 second.
2. G = 10 at W = 1.0 radian or W = 10 radians when G = 1.0

Solution: GH exclusive of amplifier 1 = $\frac{500}{100} \times 5 \times \frac{100}{350}$

$$= \frac{2500}{350} = 7.14 \text{ V/V}$$

Therefore G of amplifier 1 = $\frac{10}{7.14}$ = 1.4 V/V at W = 1.0

$$G = \frac{1}{R_iC} \qquad R_i = \frac{100 \times 250}{350} = 71.4 \text{ K ohms}$$

$$C = \frac{1}{R_iG} = \frac{1}{71.4K \times 1.4} = \frac{10^{-3}}{100} = .01 \times 10^{-3} = 10 \text{ MFD}$$

$$R_f = \frac{1}{10 \text{ MFD}} = 100K \text{ ohms}$$

If power amplifier 2 contained a time constant longer than 33 milliseconds, the system should appear as shown below:

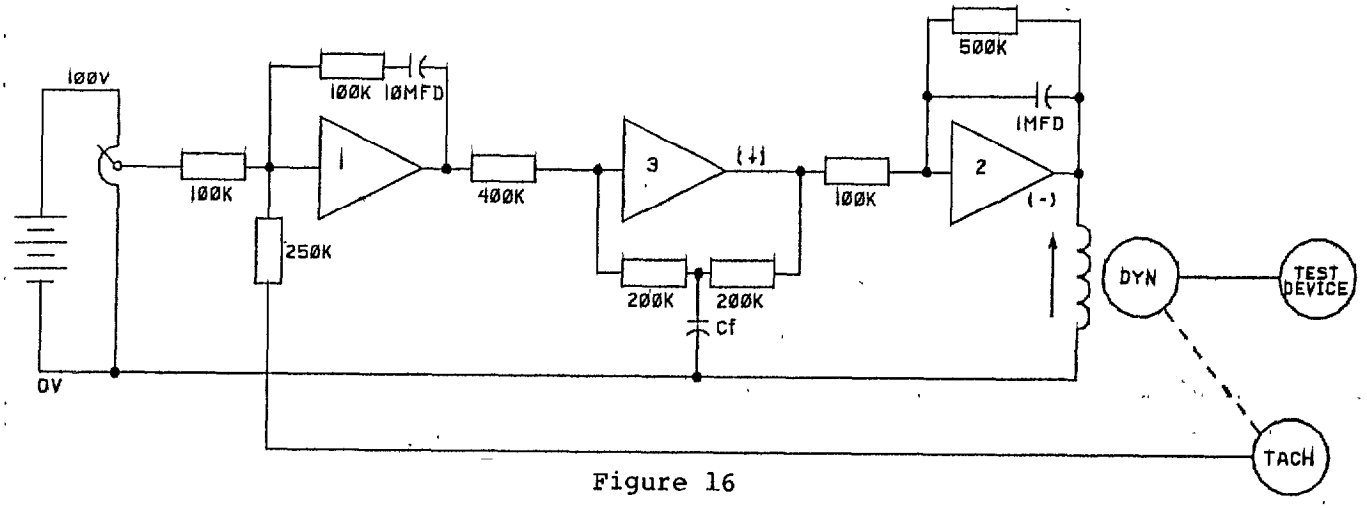


Figure 16

Time constant of amplifier 2 - 500K X 1 MFD - 1/2 second.

Therefore, the lead time constant of amplifier 3 should equal 1/2 second. Gain should remain constant at DC level.

$$\text{GAIN} = \frac{200\text{K} + 200\text{K}}{400\text{K}} = 1$$

$$\text{TIME CONSTANT} + \frac{R_1 R_2}{R_1 R_2} C_f = 100\text{K } C_f$$

$$C_f = \frac{1}{100\text{K X } 2} = 5 \text{ MFD}$$

Transient Response

Response for a perfectly compensated system as shown above should appear as shown below.

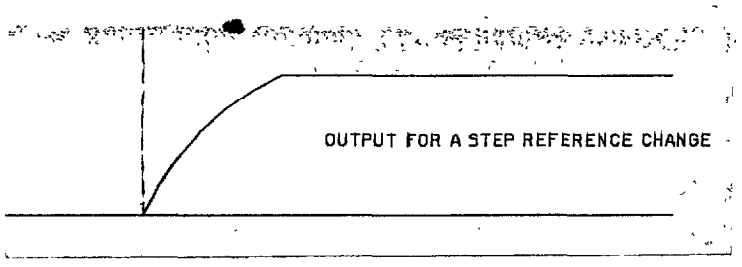


Figure 17

In a system where perfect compensation is not possible (because of changing time constants) the following guides are helpful.

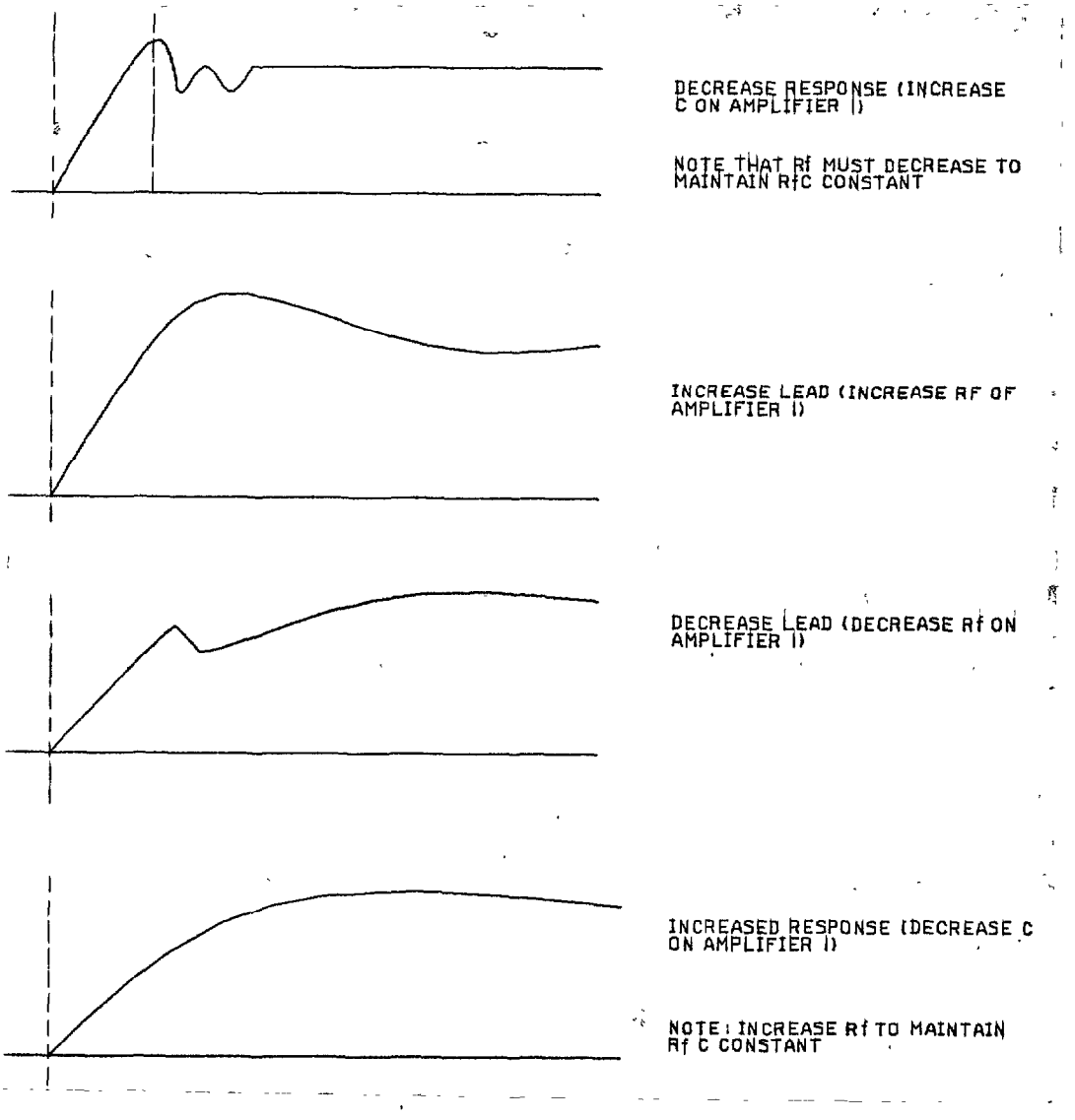


Figure 18

Saturation of any element will affect the above responses.

Operational Amplifiers

It should be recognized that the internal circuitry of the operational amplifiers is not adapted to field repair. Therefore it is recommended that no repair be attempted on the amplifier. If it is determined that the amplifier is defective, remove it and plug in a different amplifier. Return the defective amplifier to the factory for repairs.

Silicon Controlled Rectifier Exciter

This unit has been tested in the factory and has been adjusted for operation with the machines listed on the data sheet of the elementary diagram. If trouble develops or if a check is desired, follow the procedure given in the SCR manual.

Before starting the dynamometer, check the amplifier as follows:

1. Check to see that current limit pot is turned to 2.0 or higher. Throw test switch to TEST position.
2. JR4 should read 5.3 $\pm$ .3 V.
3. JR9 should read 2.5 $\pm$ .2 V.
4. SCR volts should be 25 $\pm$ .2 V.

REGULATOR ADJUSTMENT

Field Current Regulator Adjustment

Under normal conditions, the current regulator should have a response that will reach current limit in .15 to .2 seconds. (See Figure 19)

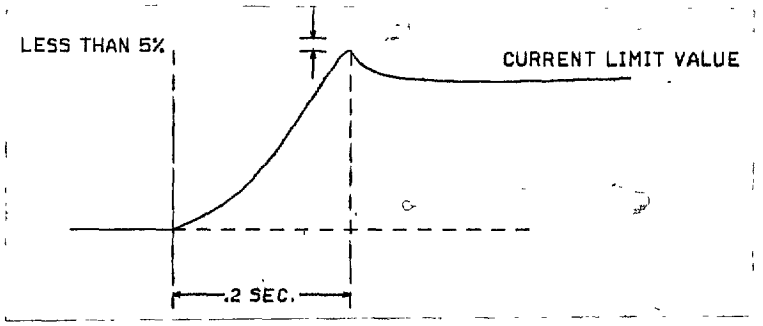


Figure 19

1. Set current limits down to 1 or 2.
2. Turn TSW switch to current.
3. Set current adjust rheostat in full CCW position (zero current).
4. Push reset button.

5. Push start pushbutton.

6. Check current response with small current steps.

7. Check scale stability here. Slowing down current response will improve scale stability.

8. Speed boost may be checked. As speed increases, field current should increase.

Current limit may need to be raised under above tests.

Speed Regulator

Response of the speed regulator should be in accordance with specifications pertaining to this drive. In the absence of specific details, the following is standard.

On 50% impact load, speed droop should not exceed 25% of top speed and recovery to 1/2% of operating speed above base speed or 1/2% of base speed below base speed shall be 8 seconds or less when the load inertia is equal to or less than dynamometer inertia.

Response will also determine the amount of scale fluctuation. In general, faster response results in greater scale fluctuation.

1. Turn TSW to speed.

2. Keep current limit set low.

3. Turn RH1 to mid position.

4. Push start pushbutton.

5. Advance speed pot slowly through the complete speed range to check stability.

6. At a speed just below base speed, put in a small speed step by rapidly twisting the speed knob. The current should not reach current limit.

7. Response should be as shown in Figure 20.

8. Adjust stability if necessary to achieve the response shown in Figure 20 (refer to notes on operational techniques).

10. Apply a torque change if possible. Response should appear as shown in Figure 21.

11. From a low value of speed, try emergency stop.

12. Advance speed and try again. Keep repeating until a stop from top speed is made successfully.

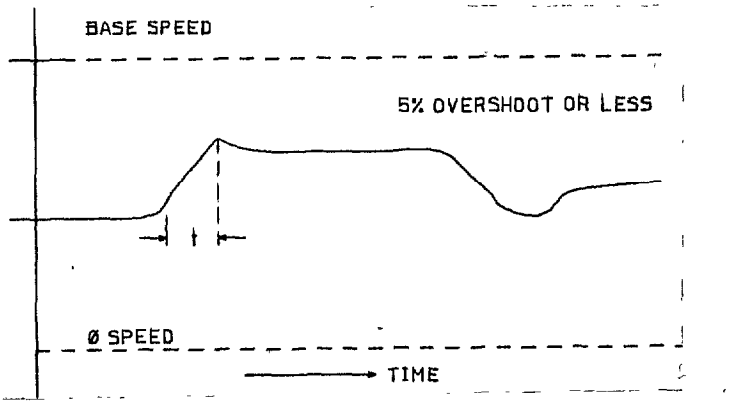


Figure 20

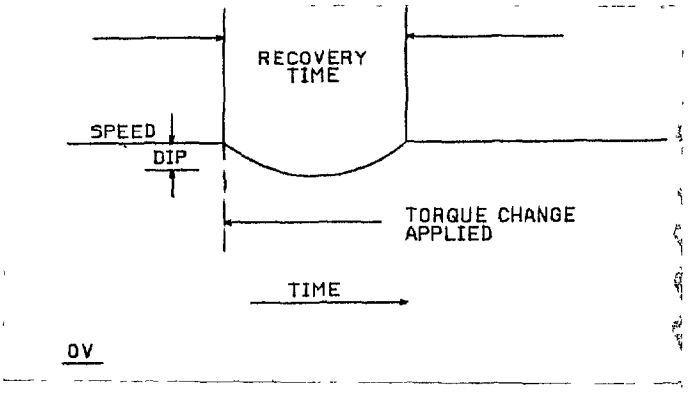


Figure 21

Speed Relay

The speed relay satisfied two functions. Overspeed detection and minimum speed dropout. It should be checked as follows:

1. Overspeed. If a DC power supply is available, disconnect tach and apply a DC power supply to SR auxiliary load. Increase power supply to 260 volts (if tach delivers 250 volts at rated top speed) and adjust pot in speed relay auxiliary card if SR does not pick up.

2. Minimum speed dropout. Decrease DC power supply to 2.5 volts and adjust pot in speed relay card if SR does not pick up.

Torque Regulator

Under normal conditions the torque regulator will have a response that will reach torque limit in 0.45 to 0.6 seconds.

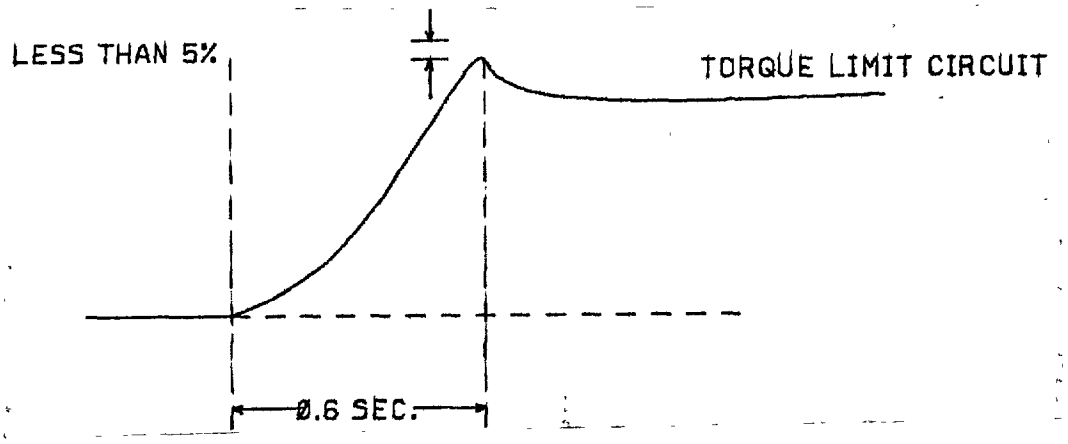


Figure 22

1. Set torque limits down to 1 or 2.
2. Turn TSW to torque.
3. Set torque adjust rheostat in full CCW position (zero torque).
4. Push reset pushbutton.
5. Push start pushbutton.
6. Check torque response with small torque steps.
7. Check scale stability. If torque response is too fast, slow it down by improving stability.
8. Check speed boost. As speed increases, torque should increase. Torque limit may need to be raised under this test.

INSTALLATION

Receiving and Unpacking

The control equipment is mounted in a rigid floor-mounted enclosure, and panels are crated and shipped in an upright position. When a panel is received, it should be handled from the top and kept upright.

A panel which is not to be installed immediately should be stored in a clean, dry location. When the panel is installed, the control should be unpacked carefully, making certain that no small parts are thrown away with the packing material.

Check the nameplate rating of the machine to be sure that it agrees with that given on the panel nameplate. The control is designed for use with specific equipment and must not be applied to any other without first checking with the nearest Apparatus Sales Office of General Electric Company.

After the panel has been unpacked, it should be handled from the removable lifting angle on top provided for that purpose. To avoid undue strains on the panel, sling the lifting angle.

Clean the inside of the control panel with a brush, soft cloth, or dry compressed air. All dirt, dust, or bits of packing material which interfere with successful operation of the panel devices should, by all means, be removed from the panel. You should likewise take care during the cleaning operation to prevent any dirt from being blown into the inaccessible spaces of the devices.

Installing the Panel

When the panel is to be installed, it is essential that it be securely fastened in a true, upright position on a level surface to permit proper operation of the devices mounted on the panel.

The enclosing case is mounted on two floor sills, one each across the back and front of the enclosing case. Floor sills consist of channel irons which are four inches deep and one and three-quarters inches high. The four-inch dimension is horizontal. On single-case installations the floor sills may be removed and the enclosing case bolted on the floor. On lineups of more than one case it is recommended that floor sills be used to give rigidity to the panel. Mounting holes for one-half-inch bolts are provided in each sill. The outline for the particular panel to be installed should be checked to determine the location of the mounting holes. It is recommended that the one-half-inch mounting bolts be grouted in the floor in the proper location for fastening the panel.

There are several possible methods of installing the floor sills. The most frequently used methods are described below:

If the panel will be installed on an existing floor, grout the one-half-inch anchor bolts in the floor and slip the panel over them. Check level position of the sills with a machinist's level, and shim the sills if necessary. Check that the surfaces of the front and back sills are in a plane, so the angle iron supports of the panel are plumb. If the panel is installed in this manner, there will be a two-inch gap between the floor and sides of the panel enclosure. This gap should be closed to prevent accumulation of dirt under the panel, which in due time may interfere with the proper operation of the panel devices.

If the concrete floor will be poured at the time that the panel is installed, grout the one-half-inch anchor bolts in the sub-floor, and slip the panel over them. Check alignment of the floor sills with a level and make sure that the panel supports are plumb. Then pour concrete around the floor sills. This should be done carefully, so as not to splash concrete on the panel devices. Care should also be taken to finish the surface of the concrete so there will be no interference of the enclosing case door and so there will be a tight joint between the floor and the enclosure side, thus preventing the entrance of dirt at the bottom of the panel.

If the concrete floor is to be poured before the panel is installed, a dummy set of floor sills can be used. These dummy sills can be made of wood of the proper size and length. The dummies should be set in place, together with the one-half-inch anchor bolts, before the concrete is poured, and then replaced with the actual sills, after the panel is received.

Provisions should be made for incoming and outgoing cable and control wires before the panel is in place. In all cases the outline for the panel being installed should be used to determine the space available, and the panel interconnection diagram should be checked for the number of wires and cables which will be installed. The cables normally enter the bottom of the panels.

Preparing the Panel for Operation

Before you can operate the panel even for a tryout without power, all devices must be placed in fully operative condition. The armatures of some of the relays and contactors may have been tied or blocked to prevent damage during shipment. Remove all of these ties and blocks before operating devices.

Scaling surfaces of unplated and laminated ac contactor and relay magnets, as well as the ends of shafts, are coated with a rust preventative. This protection against rust is a heavy grease which does not interfere with the proper operation of the devices, but which encourages the accumulation of dust and dirt which may ultimately prevent proper device operation.

Therefore, before operating the panel, you should wipe off all rust preventative. Magnet sealing surfaces may then be coated with a thin film of light maching oil.

Operate each device and relay by hand to see that the moving parts operate freely without binding. Make sure that all electrical interlocks are clean and make good contact when closed. Relays and contactors are carefully adjusted and tested at the factory. If adjustment information for individual devices is needed, however, it may be found in individual instructions for each device.

With the power leads disconnected at the machine terminals, but with all devices properly readied for operation, apply power and operate the panel to made certain that its operation is in agreement with the elementary diagram.

The leads may then be connected and the equipment put into operation.

MAINTENANCE

All electrical apparatus will provide maximum trouble-free service if given the benefit of preventative maintenance inspections and periodic cleanings. Although the frequency of the inspection periods will depend on operating conditions, it is important that a definite inspection routine and inspection schedule be followed. Again, maintenance instructions in the individual device instructions are most helpful.

Renewal Parts

Order renewal parts by part number and description, and identify the equipment for which they are required. Structural parts and complete operating devices are identified on the equipment drawing list. They are available on request.

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

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